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FORM TP 2009051

MAY/JUNE 2009

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

CHEMISTRY

Paper 02 – General Proficiency

2 hours and 30 minutes

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. **This paper consists of SEVEN questions in THREE sections.**
2. **Answer the THREE compulsory questions in Section A. Write your answers in this answer booklet.**
3. **Answer the TWO compulsory questions in Section B. Write your answers in the spaces provided in this answer booklet.**
4. **Answer ONE question from Section C. Write your answers in the space provided in this answer booklet.**
5. **Where appropriate, ALL WORKING MUST BE SHOWN in this booklet.**
6. **The use of non-programmable calculators is allowed.**

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NOTHING HAS BEEN OMITTED.

SECTION A

Answer ALL questions in this section.

Write your answers in the spaces provided in this booklet.

Do NOT spend more than 30 minutes on Question 1.

1. (a) (i) A student conducts an experiment to investigate the temperature changes that occur when various quantities of powdered zinc are added to 100 cm^3 of 0.20 mol dm^{-3} copper(II) sulphate. The apparatus used is shown in Figure 1.

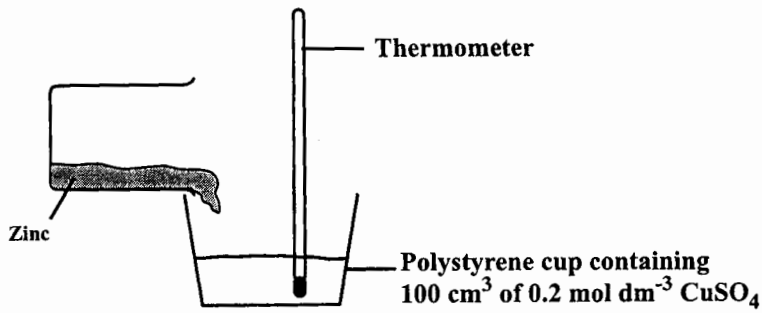


Figure 1. Apparatus used in experiment

The equation for the reaction which occurs is shown below:

$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$$

a) Describe what would be observed in the polystyrene cup in Figure 1.

_____

_____

_____

(2 marks)

b) What does the observation you described in a) above indicate about the relative positions of zinc and copper in the reactivity series?

_____

(1 mark)

c) Distinguish between an exothermic and an endothermic reaction.

_____

_____

_____

(2 marks)

(ii) Figure 2 shows the thermometer readings for the highest temperature recorded for each addition of zinc to the $CuSO_4$. The initial temperature in each case is 28°C.

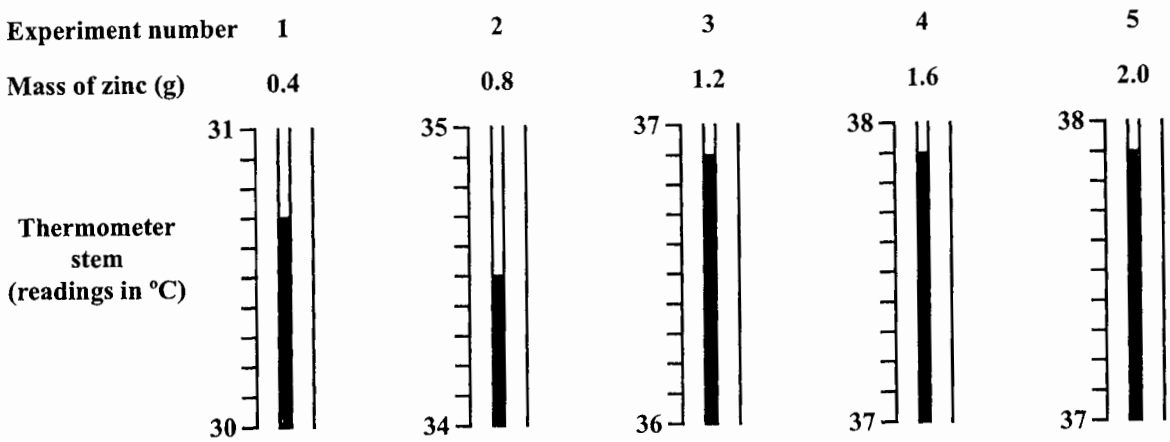


Figure 2. Final thermometer readings

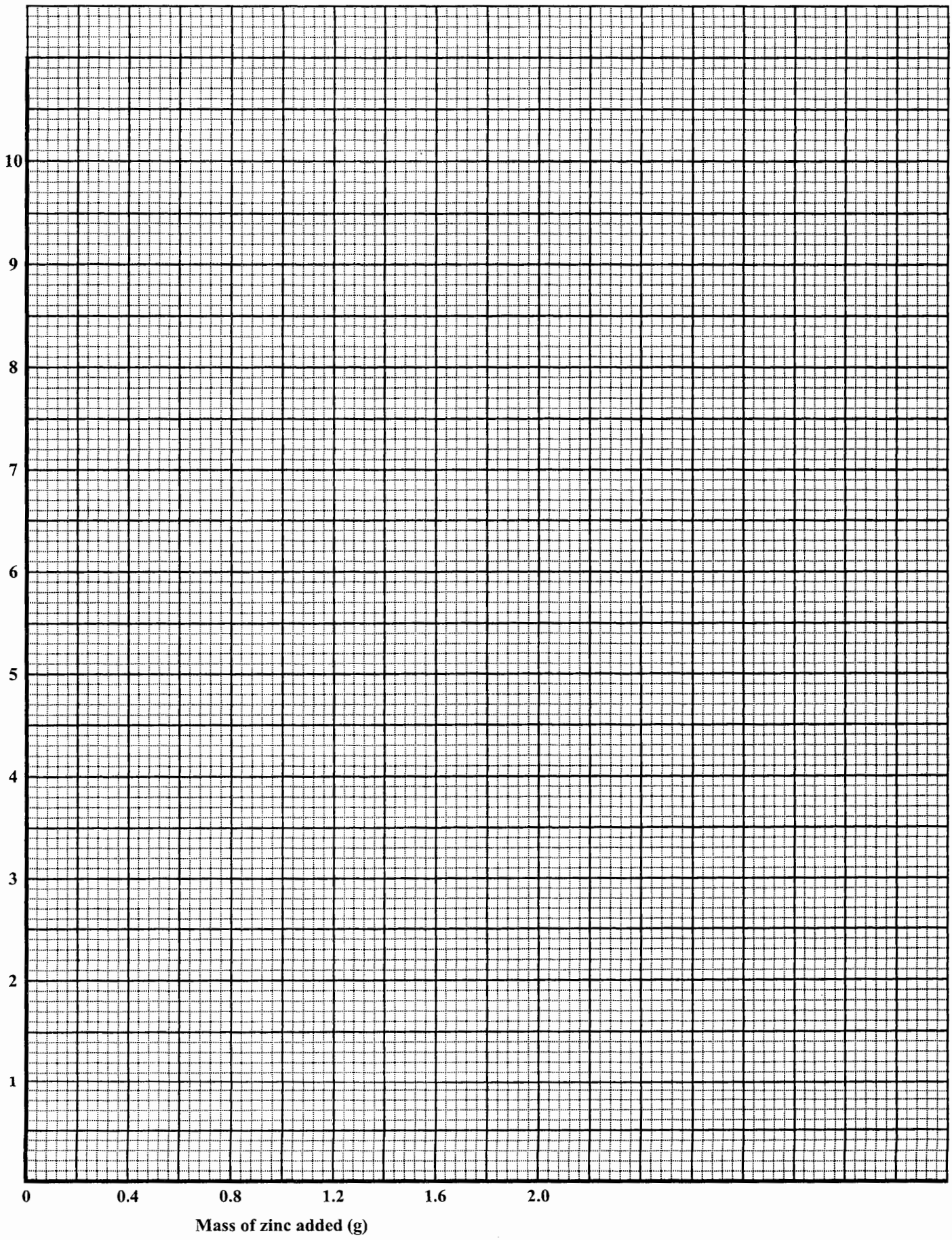
a) Using the readings in Figure 2, complete Table 1 to show the final thermometer readings, and the changes in temperature.

TABLE 1: RESULTS OF EXPERIMENT

Experiment	Mass of zinc (g)	Initial thermometer reading (°C)	Final thermometer reading (°C)	Temperature change (ΔT)
1	0.4	28		
2	0.8	28		
3	1.2	28		
4	1.6	28		
5	2.0	28	37.9	9.9

(2 marks)

b) On the graph paper provided on page 5, plot the temperature change (ΔT) against mass of zinc added. **(3 marks)**


10
9
8
7
6
5
4
3
2
1
0
0.4
0.8
1.2
1.6
2.0
Mass of zinc added (g)

c) From your graph, deduce the temperature change (ΔT) that will occur when 1.0 g of zinc is added to the copper(II) sulphate solution.

_____

(1 mark)

(iii) Calculate the minimum mass of zinc required to completely react with 100 cm^3 of 0.20 mol dm^{-3} copper(II) sulphate. Relative atomic mass of Zn = 65.

(2 marks)

(iv) Use the information in (iii) above to explain why the different masses of zinc used in Experiments 4 and 5 give approximately the same change in temperature.

_____

_____

(1 mark)

(v) Another group of students performs a similar experiment but uses silver instead of zinc. No temperature change is observed. Account for this observation.

_____

_____

(2 marks)

(b) A student conducts the following tests on a solid, R, and makes the observations recorded in Table 2.

(i) Complete Table 2 to show all of the possible inferences and write ionic equations where indicated.

TABLE 2: TESTS PERFORMED ON SOLID R

Test	Observation	Inference
A portion of R is dissolved in approximately 6 cm^3 of water. The resulting solution is divided into equal portions for Tests (1) and (2).	A colourless solution is formed.	
(1) To one portion of Solution R from above, aqueous sodium hydroxide is added dropwise until in excess, then heated.	• No precipitate is formed. • Upon heating, a pungent gas evolves which turns moist red litmus blue.	• • **(2 marks)**
(2) To another portion of R from above, dilute nitric acid, followed by silver nitrate, is added dropwise. This is followed by the addition of aqueous ammonia.	• A white precipitate is formed. • Precipitate dissolves in aqueous ammonia.	• (Ionic equation required) **(2 marks)**

(ii) Suggest a possible identity of the solid, R: _____

(1 mark)

(c) Sally, a cleaner, was preparing to clean Ms George's home. She usually cleans the kettle with vinegar and the tiles and toilet with muriatic acid (hydrochloric acid). However, the labels from the bottles containing the two cleaning agents were removed. Sally attempted to distinguish these two cleaning agents by smelling them. However, she was still unsure.

Plan and design an experiment that one can undertake in the laboratory to determine which bottle contains the vinegar and which contains the muriatic acid.

(i) Outline the procedure for the experiment.

_____

_____

_____

_____

_____

(2 marks)

(ii) List the materials necessary.

_____

_____

(1 mark)

(iii) What evidence will result from your procedure to distinguish between the two acids?

_____

_____

_____

(1 mark)

Total 25 marks

2. (a) Transitions between the three states of matter can be efficiently illustrated by using water.

Consider the information given in Figure 3 and answer the questions that follow.

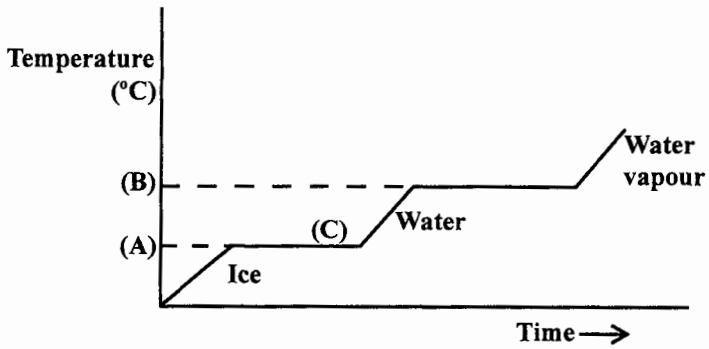


Figure 3. States of matter

(i) Name the THREE states of matter.

_____

(1 mark)

(ii) Which of the two temperatures, (A) and (B) in Figure 3 represents

melting point? _____

boiling point? _____

(2 marks)

(iii) What would be observed at (C) in Figure 3?

_____

(1 mark)

(iv) Name the process which occurs when the following conversions take place.

Water vapour → water _____

Iodine (s) → iodine (g) _____

(2 marks)

(b) Diamond and graphite are both allotropes of carbon.

(i) Explain what is meant by the term 'allotrope'.

_____

_____

(1 mark)

(ii) Which of the two allotropes of carbon, diamond or graphite, conducts electricity? Explain your answer in terms of the structure and bonding of the two allotropes.

Conducts electricity: _____

Explanation: _____

_____

_____

(2 marks)

(c) Carbon forms numerous compounds and a large number of these are hydrocarbons. These include the alkanes and alkenes. Many of these also exist as isomers.

(i) Name ONE natural source of hydrocarbons.

_____

(1 mark)

(ii) Define the term 'structural isomerism'.

_____

_____

(1 mark)

(iii) In the space provided below draw fully displayed structures of TWO isomers of a hydrocarbon with four carbon atoms. Name the TWO isomers.

Isomer #1	Isomer #2
Name:	Name:

(4 marks)

Total 15 marks

3. The first stage in the process of refining crude oil to produce more value-added products is fractional distillation.

(a) (i) Name the property of compounds upon which fractional distillation is based.

_____

(1 mark)

(ii) Name the lightest and heaviest fractions that are usually obtained from the fractional distillation of crude oil.

Lightest fraction: _____

Heaviest fraction: _____

(2 marks)

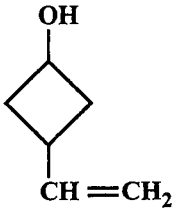
(iii) State ONE use of EACH of the fractions named in (a) (ii) above.

Lightest fraction: _____

Heaviest fraction: _____

(2 marks)

(b) Consider the structure of Compound A below and answer the questions which follow.



Compound A

(i) Draw the structure of the product formed when Compound A reacts with EACH of the following:

Bromine

(2 marks)

Sodium

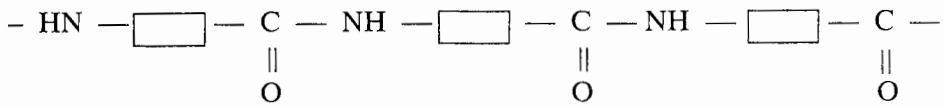
(2 marks)

(ii) Draw the partial structure of the polymer formed when Compound A undergoes ADDITION polymerisation with other molecules of A.

Use three molecules of A in the representation of the partial polymer structure.

(2 marks)

(c) B represents the partial structure of a polyamide.



Structure B

(i) Name the type of polymerisation process involved in the formation of B.

_____

(1 mark)

(ii) Polyamides of the type B can undergo both acid hydrolysis in the laboratory as well as enzyme hydrolysis in the body.

a) Draw the monomer formed when B undergoes acid hydrolysis.

(2 marks)

b) State ONE difference in the reaction conditions required for acid hydrolysis when compared to enzyme hydrolysis of polyamides.

_____

_____

(1 mark)

Total 15 marks

SECTION B

Answer ALL questions in this section.

Write your answers in the spaces provided after EACH question in this answer booklet.

4. Table 3 illustrates part of the periodic table.

TABLE 3: PART OF THE PERIODIC TABLE

PERIOD \ GROUP	I	II	III	IV	V	VI	VII	VIII
1								He
2		Be			N	O		
3	Na	Mg	Al	Si	P	S	Cl	
4				Sn				
5	Rb			Pb			I	

(a) State the basis upon which elements are arranged into groups and periods in the periodic table. Select an element from Table 3 and use its electronic configuration to illustrate your answer. **(4 marks)**

(b) Identify from Table 3:

(i) ONE element which can be extracted from its oxide by electrolytic reduction but NOT chemical reduction with coke. Give a reason for your answer. **(3 marks)**

(ii) TWO elements, one of which will displace the other from a solution containing its ions. Write an ionic equation to illustrate your answer. **(3 marks)**

(c) Iodine and chlorine will combine to form iodine monochloride (ICl). Suggest the type of bonding which would occur in this compound. Give a reason for your answer and use dot-cross diagrams to illustrate your answer. **(5 marks)**

Total 15 marks

Write your answer to Question 4 here.

Write your answer to Question 4 here.

5. (a) Ammonia is produced industrially by the Haber Process according to the following equation.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)\,;\ \Delta H = -92\ kJmol^{-1}$$

Figure 4 shows the steps in the Haber Process used for the industrial preparation of ammonia.

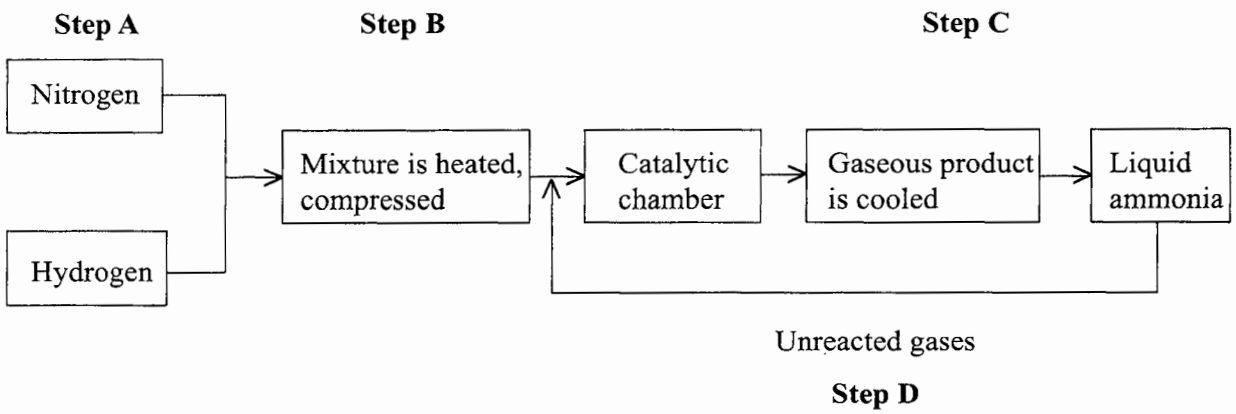


Figure 4. The industrial preparation of ammonia

(i) Name the catalyst and state the temperature used in the manufacture of ammonia. **(2 marks)**

(ii) The temperatures currently used in the manufacture of ammonia make the process cost-effective. Suggest ONE reason for this. **(2 marks)**

(iii) Suggest ONE reason for the importance of Step D in Figure 4. **(1 mark)**

(b) Ammonia can also be produced in the laboratory.

(i) Write a balanced equation to show the laboratory preparation of ammonia. **(2 marks)**

(ii) Describe a suitable test for identifying ammonia. **(2 marks)**

(c) Ammonia is used in the industrial preparation of nitric acid.
In Step 1, ammonia is heated with oxygen in the presence of a catalyst at 850 °C to produce nitrogen(II) oxide as shown in the following equation.

$$4NH_3(g) + 5O_2(g) \xrightarrow[850\ ^\circ C]{Pt - Rh\ catalyst} 4NO(g) + 6H_2O(g)$$

In Step 2, the nitrogen(II) oxide is further oxidized to form nitrogen(IV) oxide.
In Step 3, nitrogen(IV) oxide reacts with oxygen and water to produce nitric acid.

(i) Write balanced equations for Steps 2 and 3 to show the production of nitric acid from ammonia. **(4 marks)**

(ii) State TWO OTHER uses of ammonia. **(2 marks)**

Total 15 marks

Write your answer to Question 5 here.

Write your answer to Question 5 here.

SECTION C

Answer ONE question in this section.

Write your answer in the space provided at the end of this section in this answer booklet.

6. (a) The properties of clay which make it suitable for pottery production are its plasticity, porosity and its ability to be vitrified.

(i) Briefly explain what is meant by EACH of the following terms as they relate to pottery production.

- Plasticity
- Porosity
- Vitrification **(3 marks)**

(ii) Suggest ONE reason why EACH of the following makes clay a suitable substance for the making of pottery.

- Plasticity
- Porosity **(2 marks)**

(b) (i) Name ONE component of cement. **(1 mark)**

(ii) Cement is the binding agent in concrete. Identify TWO OTHER ingredients used to make concrete. **(2 marks)**

(iii) Maynard, a mason, cast a concrete floor for Mr. Smith. Before leaving, Maynard suggested to Mr. Smith that he should dampen the floor while it was hardening. From your knowledge of the components of cement and its use in making concrete, explain to Mr. Smith why it is necessary to dampen the concrete floor.

Suggest to Mr. Smith what may occur if he decides NOT to dampen the floor. **(4 marks)**

(c) Some of the ingredients in clays and cements are quite similar. Suggest if it would be appropriate to dampen clay while hardening. Give ONE reason for your answer. **(3 marks)**

Total 15 marks

7. (a) (i) What is meant by the term 'humus' as it relates to soil?

(ii) State THREE reasons why humus is an important component of soil.

(4 marks)

(b) A student puts a sample of soil in a test tube, adds sodium hydroxide and heats the mixture. For which ions is the student MOST likely testing? **(1 mark)**

(c) For years farmers in the Caribbean have been using chemicals to control many pests which attack agricultural produce.

Biological control, for example, the use of parasitic wasps to control the pink mealy bug, is another means of controlling pests.

Suggest ONE advantage and ONE disadvantage EACH of biological and chemical control of pests. **(4 marks)**

(d) (i) What is meant by the term 'hydroponics'? **(2 marks)**

(ii) Mrs. Wills is considering setting up a hydroponics farm. Advise Mrs. Wills on TWO advantages and TWO limitations of setting up such a farm.

(4 marks)

Total 15 marks

Write your answer to EITHER Question 6 OR Question 7 here.

Question Number _____

Write your answer to EITHER Question 6 OR Question 7 here.

Question Number _____

Write your answer to EITHER Question 6 OR Question 7 here.

Question Number _____

END OF TEST